

Hasnain Asif

AI Engineer · Agentic AI & LLM Systems · Full-Stack Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years of full-stack development experience, specialized in agentic AI systems, RAG pipelines, and LLM integrations deployed on AWS. Built multi-agent LangGraph pipelines, autonomous task execution systems, and enterprise-scale web platforms serving 11,000+ users across 4 countries.

TECHNICAL SKILLS

AI & LLM

Python

LangChain

LangGraph

LlamaIndex

CrewAI

OpenAI Agents SDK

Vector Databases

LLMs

Retrieval-Augmented Generation (RAG)

Transformer

ML

NLP

Prompt Engineering

Embeddings

Multi-Agent Systems

FastAPI

Cloud & DevOps

AWS Lambda

SQS

EventBridge

API Gateway

Bedrock

SageMaker

S3 / ECR

Terraform

GitHub Actions

Docker

CI/CD

Full-Stack

React

Next.js

Node.js

TypeScript

Express.js

Web3.js

Redux

GraphQL

WebSockets

Microservices

REST API

Agile

Scrum

Databases

PostgreSQL

MongoDB

ChromaDB

FAISS

Pinecone

Redis

EXPERIENCE

🕒 AI Engineer — Agentic AI & Production Systems

Self-Employed · Independent AI Engineer

Oct 2025 - Present

Designing and developing production agentic AI systems — multi-agent pipelines, RAG architectures, and LLM integrations deployed on AWS.

- Built an agentic RAG pipeline with a retrieval evaluator agent that filters documents by relevance before generation — improving answer quality in document-heavy use cases.
- Deployed a distributed multi-agent system on AWS: five Lambda functions (one per agent), S3-backed vector storage, SageMaker embeddings, API Gateway orchestration, and CloudFront + S3 for a Next.js frontend.
- Deployed IaC pipelines using Terraform and GitHub Actions for automated deployment across AWS Bedrock, Lambda, ECR, S3, and CloudFront.

Senior Software Engineer

Oct 2023 - Present

tkxel · Lahore, Pakistan

Worked on enterprise-scale systems for US-based clients, including workforce platforms and multi-tenant SaaS products.

- Led a 5-engineer team to revive a 2-year abandoned multi-tenant social platform with no documentation — owned task breakdown, PR reviews, and all technical decisions.
- Built the scheduling, shift assignment, and DST timezone engine for an enterprise workforce platform — adopted across 50+ components, serving 11,000+ users across 170+ franchises in 4 countries, with invalid assignments blocked at UI level to prevent payroll errors.
- Reverse-engineered OpenFGA authorization schema from legacy deployments — restored full role-based access control across all admin levels; separately shipped multi-currency Stripe subscriptions (PKR, USD, GBP) on a platform designed for 1M+ audience members across 1,000–5,000 brand Hubs.

Software Engineer

May 2022 - Oct 2023

tkxel · Lahore, Pakistan

Built full-stack features for enterprise tools and internal platforms.

- Delivered enterprise helpdesk features for Tikit, an MS Teams-integrated ticketing platform for high-volume IT, HR, and Finance workflows.
- Developed a Custom Views system for an MS Teams-integrated enterprise helpdesk — enabled persistent multi-filter configurations across thousands of concurrent tickets.

Software Engineer

Oct 2021 - Jan 2022

Invozone · Lahore, Pakistan

Frontend-led delivery on a multi-chain DEX product for a Web3 startup — cross-chain token swaps across four networks using Web3 infrastructure.

- Directed frontend delivery on a multi-chain DEX across Ethereum, BSC, Polygon, and Avalanche — integrated MetaMask and Coinbase wallets via Web3.js.

Software Engineer

Aug 2020 - Oct 2021

OptimusFox · Lahore, Pakistan

Built full-stack blockchain-based platforms across fintech and civic tech — led delivery on multiple products.

- Served as tech lead on a token offering (ITO) platform — designed ERD, owned full-stack architecture, and routed tasks across a 4-engineer team.
- Built a blockchain-based tamper-resistant e-voting system with three role-based panels — admin, candidate, and voter — with on-chain ballot execution via Web3.js.

PROJECTS

DocInsight — Verified Document Q&A

LangGraph · LangChain · ChromaDB · OpenAI · FastAPI

A multi-agent RAG system that eliminates manual document review — three specialized agents gate relevance, generate grounded answers, and verify factual support before any response reaches the user. Targets the 1.8 hours/day that knowledge workers lose to unverified information retrieval.

- Built a 3-agent LangGraph pipeline with OpenAI — Relevance Checker, Research Agent, and Verification Agent — answers verified against source passages before being returned to the user.
- Implemented hybrid retrieval combining BM25 (40%) and ChromaDB vector search (60%) — improves recall on exact-match queries that pure semantic search consistently misses.
- Designed conditional routing so failed verification triggers automatic re-research — unverified answers never surface to the user.

Sidekick — Autonomous AI Work Agent

LangGraph · OpenAI · Playwright · Serper API · Python REPL

A two-model agentic system that executes multi-step tasks across browser, code, files, and search without human supervision — a structurally independent evaluator verifies success criteria are met before any result is returned. Addresses the 62% of workday spent on tasks that could run unattended and the 9.5 minutes of focus lost each time a worker checks in on a running task.

- Built a two-model LangGraph pipeline — worker executes across browser automation, Python REPL, file I/O, and web search, while a separate evaluator independently verifies task completion, eliminating self-evaluation bias.
- Implemented typed feedback loop — structured feedback is injected into the worker system prompt on retry, so the agent knows exactly why it failed before reattempting.
- Built a human-in-the-loop escalation mechanism — when the agent repeats the same mistake after feedback, it surfaces the blocker to the user instead of looping silently.

EDUCATION

BS Software Engineering

University of Engineering & Technology, Taxila

Rawalpindi, Pakistan

CERTIFICATIONS

- IBM RAG and Agentic AI Specialization
Coursera · IBM
- AI Engineer Production Track: Deploy LLMs & Agents at Scale
Udemy · Ed Donner
- Agentic Track: The Complete Agent & MCP Course
Udemy · Ed Donner
- Serverless Framework Bootcamp: Node.js, AWS & Microservices
Udemy · Ariel Weinberger
- Data Science Internship Certificate
Skilled Score · Gufhtugu Publications